

Math 421

Monday, September 22

Announcements

- Homework 3 due Friday
- Grace's Drop-In Hours
 - Wednesday: 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4

Chapter 1: Introduction

Section 4: The Completeness Axiom

Maximum & Minimum

Def. Let S be a nonempty subset of $\mathbb{R}$. If S contains a *largest element*, M_0 , then we call M_0 the **maximum of S** and write $M_0 = \max S$.

Examples

Upper & Lower Bounds

Def. Let S be a non-empty subset of $\mathbb{R}$. If a real number M satisfies $s \leq M$ for all $s \in S$, then M is an **upper bound** for S and the set S is **bounded above**.

Bounded

Def. Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be **bounded** if it is bounded above and bounded below.

Activity

For each of the following sets find (if they exist) the maximum, minimum, an upper bound, and a lower bound

1. $\mathbb{Z}$
2. $\mathbb{Q}$
3. $\mathbb{N}$
4. $(a, b]$
5. $\{r \in \mathbb{Q} : 0 \leq r \leq \sqrt{2}\}$
6. $\{\frac{1}{n} : n \in \mathbb{N}\}$

Supremum & Infimum

Def. Let S be a nonempty subset of $\mathbb{R}$. If S is bounded above and S has a *least upper bound*, then we call it the **supremum of S** and denote it by $\sup S$.

Group Activity/Discussion

- ❖ Give an example of the following:
 - A bounded set that has a minimum value but no maximum value.
 - A set that has no least upper bound (or its least upper bound is infinite).
- ❖ When is it true that $\sup S \in S$ and $\inf S \in S$?
- ❖ Are $\sup S$ and $\inf S$ unique?

Activity

Find the supremum/infimum of the following sets (if they exist).

1. $\{n \in \mathbb{N} : n \text{ is prime and } n < 10\}$

2. $\{n^2 : n \in \mathbb{N}\}$

3. $\left\{\sin \frac{n\pi}{4} : n \in \mathbb{N}\right\}$

4. $\{x^3 : x > 3\}$

5. $\{n \in \mathbb{N} : 2 < n < 3\}$

Equivalent Definition

Thm. Let $s \in \mathbb{R}$ be an upper bound for a set $A \subseteq \mathbb{R}$. Then $s = \sup A$ if and only if for every $\varepsilon > 0$ there exists some $a \in A$ with $s - \varepsilon < a$.

Discussion Questions:

- ❖ Describe what this theorem is saying in your own words.
- ❖ Convince yourselves that this theorem is true, it might be helpful to think in terms of a specific example and then see how it can generalize.

Completeness Axiom

Completeness Axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound. In other words, $\sup S$ exists and is a real number.

Q: Does the completeness axiom hold for $\mathbb{Q}$?

Corollary: Every nonempty subset S of $\mathbb{R}$ that is bounded below has a greatest lower bound.

Archimedean Property

Thm. If $a > 0$ and $b > 0$, then for some positive integer n , we have $na > b$.

Discussion Questions:

- ❖ What is this theorem saying in your own words?
- ❖ What does this property say if $a = 1$?
- ❖ What does this property say if $b = 1$?

Archimedean Property

Discussion.

Archimedean Property

Proof.

Density of $\mathbb{Q}$

Thm. If $a, b \in \mathbb{R}$ and $a < b$, then there is a rational $r \in \mathbb{Q}$ such that $a < r < b$.

Discussion Questions

- ❖ Is the empty set bounded? Does it have a least upper bound? A greatest lower bound?
- ❖ What does it mean for a set to be **unbounded**?

The Symbols $+\infty$ & $-\infty$